Circuit Lab B/C Condensed Key

January 10, 2026

Greenbrier Invitational

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KEY KEY KEY KEY KEY KEY KEY KEY

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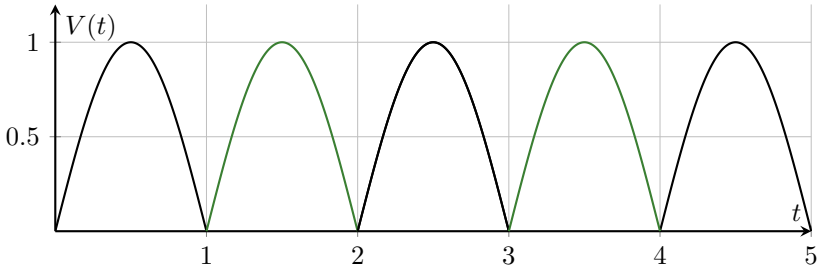
1. Do **NOT** write on this packet. Only answer on your answer sheet.
2. The points per question are boxed in the margin next to the question.
3. You may separate the test, but ensure you put it back in order.
4. You may remove materials from your binder ***AFTER*** the time has started.
5. **SHOW ALL OF YOUR WORK**
6. Tiebreaker questions have TB#(order) in front of them. (QN6, QN18, QN22)

Multiple Choice [2 each]

Q1. A	Q2. B	Q3. D	Q4. D	Q5. A	Q6. C
Q7. B	Q8. B	Q9. B	Q10. A	Q11. C	Q12. A
Q13. D	Q14. C	Q15. C	Q16. C	Q17. C	Q18. B

Free Response [24 pts]

Q19. [2]



Q20. [2]

Electron current is the flow of electricity from the negative to the positive terminal [1]. It occurs at very slow speeds **OR** any in the range from 10^{-3} to $10^{-4}\ m/s$. [1]

Q21. [5]

Energy flux refers to the rate of energy flow through a surface [1]. It travels as an electromagnetic wave next to the conductor [1], entering the resistive load to do work [1]. It is measured in W/m^2 [1], often described using a Poynting Vector [1]

Q22. [3]

Heat/Thermal energy [1], Kinetic Energy [1], Sound/Sonic Energy [1]

Q23. [2]

$28.3\,\Omega$ [1]

Q24. [4]

$V_C(t) = V_{max}(1 - e^{-t/\tau_{RC}})$ [1]
$0.47 = 1 - e^{-t/\tau_{RC}}, e^{-t/\tau_{RC}} = 0.53$ [1]
$-t/\tau_{RC} = \ln(0.53), t/\tau_{RC} = -\ln(0.53)$ [1] ≈ 0.634 [1]

Q25. [8]

$V_D = V_E = V_B$ [1]

$V_G = V_C = V_F$ [1]

$$\frac{1}{\frac{1}{R_{AD}} + \frac{1}{R_{AE}} + \frac{1}{R_{AB}}} \boxed{1} + \frac{1}{\frac{1}{R_{HD}} + \frac{1}{R_{DC}} + \frac{1}{R_{HE}} + \frac{1}{R_{EF}} + \frac{1}{R_{BF}} + \frac{1}{R_{BC}}} \boxed{3} + \frac{1}{\frac{1}{R_{GH}} + \frac{1}{R_{GC}} + \frac{1}{R_{GF}}} \boxed{1}$$

If all $R = 1$, $R_{eqv\ AG} =$ 0.833Ω [1]

____/ 11__ RUBRIC (25% of overall score)

____/ 1__ LEDs ARRANGEMENT

Series (0 points)
Parallel (1 points)

____/ 10__ SOLUTIONS

SOLUTION 1
555 timer in astable mode

- R1 = 10kΩ
- R2 = 100kΩ
- C = 10μF
    - T = 2.5s

555 timer output goes into Decade Counter, Decade Counter output

- **Q0:** Green + Resistor
- **Q1:** Yellow + Resistor
- **Q2:** Red + Resistor LEDs
- **Q3:** Reset Pin

SOLUTION 2

- Three NPN transistors in series
- Each base driven by an RC charging network, Each RC ≈ 2–3 s
- Each transistor powers one LED
- Capacitors charge sequentially
- Final capacitor discharges through diode to reset first stage

SOLUTION 3

- 555 astable (~2.5 s)
- Output feeds RC ladder
- Voltage thresholds trigger transistors sequentially
- Each transistor controls one LED
- Discharge path resets ladder

SOLUTION 4

- Switch gives clock pulses
- Each press advances decade counter
- Q0/Q1/Q2 to drive LEDs
- Q3 to reset
- Timing controlled by human (Question did not specify automated system)

Greenbrier Invitational - Repaired practice edition - Answer key

Division C | Circuit Lab | Repair appendix

This appendix contains authored replacements and supplied written alternatives. It does not reconstruct unpublished historical material. For the listed item IDs, use this appendix instead of the archived wording and key.

ANSWER KEY - Review only after submitting.

1. (2 points) Authored replacement for source item Question 1: A negative charge's force is…

Unit: Electrostatics, historical ideas, and event reasoning.
Worked-case data: Two point charges keep their values while separation triples. Predict force.

A. Opposite the electric field

B. Always with the field

C. Always zero

D. Independent of its charge

Correct: A

6. (2 points) Authored replacement for source item Question 6: Define "Potential" in the context of Circuit Lab.

answer: Electric potential energy per charge. (2 points).

repair-case. (6 points) Authored written substitute for unavailable source material: Solve this case and explain your reasoning: Two point charges keep their values while separation triples. Predict force.

reasoning: Use F proportional to 1/r^2.
Replacing r with 3r multiplies the denominator by nine.
The new magnitude is F/9; attraction or repulsion is unchanged. (3 points).

conclusion: Direction depends on signs, not distance alone. (3 points).